

Designing a near-optimal isolated microgrid using active learning

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ABSTRACT

This paper proposes an innovative active-learning-based framework to identify near-optimal groupings of an islanded microgrid network. We aim to maximize service capacity and minimize total active power losses while maintaining voltage and frequency within operational constraints. Instead of exhaustive optimization, we adopt active learning to approximate optimal configurations, significantly reducing computational costs efficiently. The methodology involves designing a solution using machine learning techniques, implementing the solution computationally, and evaluating the benefits and limitations of the proposed approach. The results show that active search method efficiently identified the near-optimal microgrid grouping after evaluating only three configurations using both k-nearest neighbour regression and Gaussian process regression techniques. The method demonstrated superior computational efficiency while ensuring compliance with operational constraints.

1. Introduction

Microgrid systems represent an approach to decentralized energy generation, distribution, and consumption. As the global energy landscape shifts towards renewable sources and grid resilience, the role of microgrids becomes increasingly prominent [1]. These systems help optimize energy management, reduce greenhouse gas emissions, and facilitate the integration of renewable resources into the electricity grid. However, achieving these objectives necessitates sophisticated control strategies that adapt to complex, often unpredictable environments [1].

As for the islanding process, this paper focuses on identifying an islanding configuration to extract the most from the existing sources. To this end, renewable sources may be employed [2], along with dispatchable generation. The strategy focused here seeks the best single configuration following a disconnection from the grid. It differs from [3], where a dynamic island forming is proposed.

Such a methodology allows the configuration to change over time. The strategy proposed in this paper may emulate this characteristic if each period of interest is considered as an islanding interval.

In real-world applications, absolute optimality is often impractical due to computational constraints. This study employs active learning techniques to achieve near-optimal solutions for efficiently clustering islanded microgrids. By iteratively selecting promising configurations through machine learning predictions, the method rapidly converges

to high-performance solutions, balancing optimality and computational feasibility.

In this context, this paper proposes a machine learning approach that uses active learning to define a near-optimal islanded microgrid clustering aimed at maximizing service capacity. The term active is used herein because the search method interacts with the analyst by indicating the objects subjected to the direct analysis procedure. This concept bears similarity with active learning techniques [4], which have been studied within the scope of classification [5,6] and multivariate calibration tasks [7]. However, such active learning techniques are aimed at the construction of classification or regression models. In contrast, the present work concerns an optimization problem, which does not necessarily involve constructing an explicit model.

The regression techniques applied here are a k-nearest-neighbour method [8] and a Gaussian process [9]. The algorithm searches for the most promising object with the best features for building a model. It was applied to a modified IEEE 37-node test feeder in order to define the best clustering for the network under islanded mode.

The contributions of this paper are:

- Proposing an intelligent algorithm to define a near-optimal clustering for an islanded microgrid to maximize service capacity.

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- Presenting a structured methodology for searching for optimal or near-optimal values based on the active machine learning paradigm.
- Presenting an active search algorithm using regression techniques that converge to the near-optimal faster than other studies.

The remainder of the paper is structured as follows. The Section 2 presents a brief review of microgrid operation in islanded mode. The Section 3 describes the proposed search method based on active machine learning. The Section 4 describes simulation results and discussions. In the conclusion, the final considerations of this research are presented.

2. Microgrid operation in islanded mode

2.1. Inverter operation mode

This section discusses different ways of operating a microgrid in isolated mode. Distributed generators are connected to the grid by an inverter-based interface [10]. This paper assumes two possible control modes for the operation of the inverter during the islanded condition, given by PQ control mode and Voltage Source Inverter (VSI) control mode.

Renewable energy sources such as photovoltaic systems are controlled by PQ mode and, hence, are considered as negative power loads. The VSI mode is used for dispatchable generators (such as microturbines). Since the frequency and voltage are dependent upon the demand, the active and reactive powers generated at the unit (P_{gk} and Q_{gk} , respectively) are given by droop control equations, as depicted in Eq. (1) and Eq. (2) [11]:

$$P_{gk} = \frac{f_{ref} - f}{n_k} \quad (1)$$

$$Q_{gk} = \frac{V_{kref} - V_k}{m_k} \quad (2)$$

where f is the system's frequency and f_{ref} is the frequency at no load; V_k is the terminal voltage, whereas V_{kref} is the voltage at no load; m_k and n_k are the droop coefficients that represent the share of active and reactive powers of the inverter in bus k .

2.2. Power flow in islanded microgrid

The power flow algorithm formulation for islanded microgrids must take into consideration the operating characteristics of these systems [12,13]: (1) the lack of a swing bus; which implies that the system frequency modifies at each update of the operating point; (2) the low X/R ratio (where X and R stand for the line reactance and resistance, respectively); which makes the P/θ and Q/V couplings no longer guaranteed, therefore compromising the Newton–Raphson method.

Within this context, the authors in [12] propose a power flow methodology. The modified power flow problem considers the following in its formulation: The variable vector is given by Eq. (3).

$$x = [\theta^T \mathbf{V}^T f V_1]^T \quad (3)$$

where θ is the vector of voltage angles; $\mathbf{V}$ is the vector of voltage magnitudes, except Bus 1, which has the voltage's magnitude represented by V_1 and is the only one with the defined angle; f is the frequency of the system, which is declared as an additional variable due to the absence of the swing bus. Conventionally, at the initialization of the iterative process, the system voltages are set to $1 < 0$ p.u., and f is assigned 1 p.u. The vector with the power mismatch equations is given by the Eq. (4).

$$\Delta' = [\Delta \mathbf{P}^T \Delta \mathbf{V}^T \Delta P_{sys} \Delta Q_{sys}]^T \quad (4)$$

For a particular bus k , ΔP_k and ΔQ_k are defined by Eq. (5) and Eq. (6), respectively.

$$\Delta P_k = P_{gk} - P_{loadk} - P_{calc_k} \quad (5)$$

$$\Delta Q_k = Q_{gk} - Q_{loadk} - Q_{calc_k} \quad (6)$$

where P_{loadk} is the total active power demand and Q_{loadk} is the total reactive load. P_{gk} and Q_{gk} are given by (1) and (2). Vectors $\mathbf{P}_k$ and $\mathbf{Q}_k$ contain the scheduled values for the active and reactive powers for a particular bus k ; vectors $\mathbf{P}_{calc_k}$ and $\mathbf{Q}_{calc_k}$ contain the calculated values for the active and reactive powers, which are described by the conventional power flow equations. ΔP_{sys} and ΔQ_{sys} are given by Eq. (7) and Eq. (8), respectively.

$$\Delta P_{sys} = \sum_{k=1}^d P_{loadk} + P_{loss} - \sum_{k=1}^d P_{gk} \quad (7)$$

$$\Delta Q_{sys} = \sum_{k=1}^d Q_{loadk} + Q_{loss} - \sum_{k=1}^d Q_{gk} \quad (8)$$

where P_{loss} is the system's active power loss, Q_{loss} is the reactive power loss, and d is the number of dispatchable generators.

The Jacobian matrix for the system of power mismatch equations given by (4) corresponds to Eq. (9) and the variable increments Δx are calculated using the Levenberg–Marquardt method [13] given by Eq. (10).

$$\mathbf{J}' = \begin{bmatrix} \frac{\partial \mathbf{P}}{\partial \theta} & \frac{\partial \mathbf{P}}{\partial \mathbf{V}} & \frac{\partial \mathbf{P}}{\partial f} & \frac{\partial \mathbf{P}}{\partial V_1} \\ \frac{\partial \mathbf{Q}}{\partial \theta} & \frac{\partial \mathbf{Q}}{\partial \mathbf{V}} & \frac{\partial \mathbf{Q}}{\partial f} & \frac{\partial \mathbf{Q}}{\partial V_1} \\ \frac{\partial P_{sys}}{\partial \theta} & \frac{\partial P_{sys}}{\partial \mathbf{V}} & \frac{\partial P_{sys}}{\partial f} & \frac{\partial P_{sys}}{\partial V_1} \\ \frac{\partial Q_{sys}}{\partial \theta} & \frac{\partial Q_{sys}}{\partial \mathbf{V}} & \frac{\partial Q_{sys}}{\partial f} & \frac{\partial Q_{sys}}{\partial V_1} \end{bmatrix} \quad (9)$$

$$\Delta x = \frac{\mathbf{J}'^T \mathbf{J}' + \lambda \mathbf{I}}{\mathbf{J}'^T \Delta'} \quad (10)$$

where λ is a damping factor, and $\mathbf{I}$ is the eye matrix. This technique is applied to overcome convergence problems in the Newton–Raphson. Finally, for the $i+1$ th iteration, the variable vector is calculated by the Eq. (11).

$$\mathbf{x}^{i+1} = \mathbf{x}^i + \Delta x \quad (11)$$

This method efficiently avoids convergence problems [14]. The convergence of the Newton–Raphson method results in the state of the microgrid for a specific operating scenario and the performance of primary control, i.e., the generators' droop response.

3. Proposed methodology

The proposed methodology is grounded in the active machine learning paradigm, specifically active search, which aims to identify high-performing solutions with a limited number of evaluations. The active learning strategy employed does not perform an exhaustive search of the entire solution space, hence it does not strictly guarantee global optimality. Instead, the methodology iteratively approximates optimal solutions by evaluating only the most informative configurations suggested by surrogate models, resulting in efficient convergence to practically near-optimal outcomes.

3.1. Problem description

In this article, the concept of microgrid clustering is applied in the context of islanded operation. The concept of microgrid clustering consists of dividing a system into smaller subsystems. In this context, the microgrid is divided into multiple islands that can be grouped, forming different clusters.

It is assumed that a microgrid can be divided into smaller units, referred to as submicrogrids, as depicted in Fig. 1, which shows a

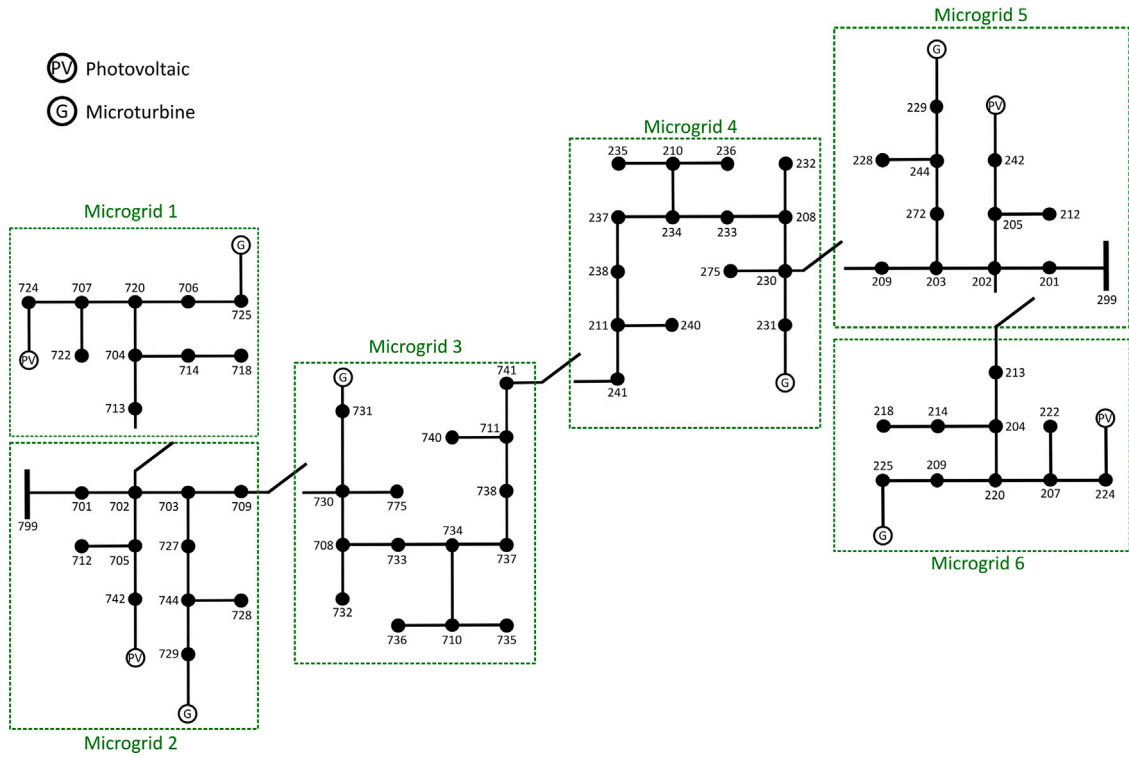


Fig. 1. Microgrid isolated from the main system and divided into six smaller microgrids.

modified IEEE 37-Bus test system with distributed generators (photovoltaic systems and microturbines), subdivided into six submicrogrids. These smaller microgrids can be grouped into different arrangements, each referred to as a cluster, where microgrids 1 to 6 are represented by M1, M2, M3, M4, M5, and M6, respectively. Multiple clusters can be organized into different groupings. Table 1 shows the 32 possible groupings derived from the submicrogrids in Fig. 1.

In this work, it is considered that the time of disconnection from the main grid and the amount of time that the system will stay disconnected are known. From a planning perspective, the proposed methodology aims to determine the grouping that, for a specified period of disconnection, has the maximum service capacity. This is the grouping where all clusters maintain the frequency and voltage levels within operational limits and minimizes total losses. One strategy to assess the service capacity of the different groupings is to perform a power flow analysis over the specific period of disconnection, for each possible cluster. Note that, in the study, both load and photovoltaic generation profiles during the disconnection hours are considered, and the microturbine output follows the droop control (Eqs. (1) and (2)). Thus, the methodology takes advantage of primary control, which requires no communication among the generating units.

The strategy previously described is associated with high computational cost, as the number of clusters requiring power flow analysis increases with the penetration of distributed generation and the geographical extent of the microgrid. In this context, as part of the planning process, this work proposes an intelligent algorithm based on active search to define the near-optimal microgrid clustering for islanded operation. The AI-based methodology eliminates the need to perform power flow analysis for all possible clusters; instead, the analysis is limited to a reduced set of promising groupings.

For this purpose, this article proposes a novel methodological framework for microgrids that leverages machine learning to determine the near-optimal grouping of an islanded microgrid network, aiming to maximize service capacity during islanded operation while satisfying operational constraints such as voltage and frequency limits.

Operational constraints — frequency within [59.5 Hz, 60 Hz] and voltage within acceptable IEEE standard ranges (0.95–1.05 p.u.) — are explicitly evaluated at each iteration through detailed power flow simulations. Surrogate models predict active power losses to select promising candidates, but the feasibility check (compliance with operational constraints) is performed rigorously via these simulations. Any candidate violating these constraints is deemed infeasible, and thus excluded from the selection as a near-optimal solution.

In the context of this study, a near-optimal grouping refers to a configuration of islanded microgrids partitioned into clusters that operate in coordination to maximize service capacity and/or minimize total active power losses, while satisfying system constraints such as voltage and frequency limits. Each grouping represents a unique combination of microgrids that are temporarily interconnected to form one or more autonomous clusters during the disconnection period. Optimality is evaluated based on the grouping's ability to supply loads, ensure operational stability, and reduce electrical losses in the network. Therefore, “near-optimal grouping” is not a topological or graph-theoretical concept in this context, but rather a practical operational strategy derived from the evaluation and comparison of multiple possible combinations of microgrid clusters under power system constraints.

3.2. Objective function

In this study, the objective function y represents the total active power loss in a given microgrid grouping during islanded operation, which must be minimized. The loss is computed as the sum of the losses in each cluster that forms the group, based on power flow calculation.

Operational constraints were defined clearly to reflect realistic islanded conditions: frequency constraints ([59.5 Hz, 60 Hz]) explicitly target the most common islanded scenario—frequency drops due to generation deficits following grid disconnection. IEEE 1547-2018 [15] specifies broader ranges, including transient overfrequency limits (up to approximately 60.5 Hz), but this study emphasizes underfrequency conditions common in isolated operational scenarios. Similarly, voltage constraints are defined explicitly within $\pm 5\%$ of nominal values

(0.95–1.05 p.u.) as commonly accepted by IEEE standards to ensure safe and reliable microgrid operation.

In this study, the optimization objective function minimizes total active power losses for a given microgrid grouping during islanded operation. Operational constraints, namely frequency and voltage limits, are explicitly evaluated via power flow simulations for each candidate grouping. A grouping is considered feasible only if it maintains frequency within [59.5 Hz, 60 Hz] and voltage within $\pm 5\%$ of nominal voltage (0.95–1.05 p.u.) at all buses. Formally, the problem is given by Eq. (12) and Eq. (13):

$$\min_{g \in G} y(g) = \sum_{i=1}^{n_C} \text{Loss}(C_i) \quad (12)$$

subject to:

$$59.5 \leq f_i \leq 60.0 \quad \forall i, \quad 0.95 \leq V_i \leq 1.05 \quad \forall i \quad (13)$$

where $g \in G$ is a grouping of microgrids, C_i are the clusters in group g , n_C is the total number of cluster of a grouping, and f_i , V_i are the frequency and voltage of cluster i , respectively.

The optimization is performed over a finite set of groupings, with the property $y(g)$ estimated through surrogate models and validated through power flow simulation.

3.3. Active learning

In many analytical applications, the problem consists of finding an object with an optimal or near-optimal value for a specific y -property of interest within a given pool of objects [16].

Active learning aims to select a reduced and non-redundant pair to produce a training set with information that faithfully represents an unlabelled dataset [17]. An active search is essentially an optimization strategy based on active machine learning, in which new experimental data is selected iteratively to find an experiment that optimizes the objective. It is considered an area of active machine learning and it is a method proposed in the literature to find the optimal of objective functions that are costly to evaluate. Besides, this technique aims to minimize the number of objective function evaluations. It applies when there is no closed expression for the objective function, but observations of that function can be obtained from the sampled values.

This system applies an active search method to find objects with optimal or near-optimal y -values based on c -components obtained through indirect and less costly methods. The search algorithm progresses sequentially, starting from a small subset of objects with known y -values. At each iteration, a multivariate weighted-averaging technique — k -nearest neighbour regression (kNNR) or Gaussian process regression (GPR) — is employed to estimate $\hat{y}$ -values for objects with unknown y -values. The object with the best $\hat{y}$ -value is selected as the most likely candidate for optimality. This candidate is then subjected to a direct analysis procedure to evaluate the y -property.

3.3.1. Kennard–Stone algorithm

The initialization procedure is carried out using the Kennard–Stone (KS) algorithm, which is a classical method for selecting objects in a near-uniform manner based on the Euclidean distances between each pair of x -vectors [18]. This method is initialized by selecting n_0 objects based on the x -vectors alone, i.e., without using any information concerning the associated y -values. The first two selected objects are separated by the most considerable distance in the x -space. Each subsequent object is chosen according to a maximum–minimum procedure to avoid selecting objects that are close together.

The KS algorithm selects an initial subset of microgrid configurations with maximum geometric dispersion in the input space. The first two points selected are those with the largest Euclidean distance between them, and subsequent points are added iteratively to maximize the minimum distance to already-selected points. This ensures that the initial sample is representative of the diversity of the entire solution space, which is crucial for training effective surrogate models in the early stages of the active search.

3.3.2. K -nearest neighbour regression

The method kNNR is a type of machine learning algorithm that is robust, simple to implement, and computationally inexpensive [19,20]. The number of neighbours to interpolate over, k , is user-specified, and the k training points with the smallest Euclidean distance in the input feature space are defined as nearest. Thus, kNNR can predict an output value by taking a weighted average of data points with similar input features.

Algorithm 1 gives a pseudo-code description of our approach, using kNNR. In this algorithm, $\mathbf{X}$ is the matrix with N objects of the data set, y is the vector with the y -value for the selected objects, i_{sel} is the vector with n_0 indexes selected by the KS algorithm, n_0 and n_{max} is the initial number and the maximal number of objects, respectively. $\mathbf{D}$ contains the Euclidean distance matrix. The only parameters that need to be chosen in the proposed algorithm are the total number $n_{max} \in [n_0, N]$ of objects to be selected and the number k of nearest neighbours.

The kNNR algorithm receives $\mathbf{X}$, y , $freq$, $\mathbf{D}$, i_{sel} , i_{sel}^- , n_0 and n_{max} ; and returns i_{sel} that contains n_{max} objects selected. The $N_k(k, i_{sel})$ is the index set of the k -nearest-neighbours of an object x_i (in the sense of smallest Euclidean distance) among the objects already selected. In the case studies, different k values were used to investigate this parameter's effects on active search [16]. The results were compared with a simpler version of the kNNR technique, in which all objects already selected are used in the estimates of y . In this case, it is considered that $k = n$ for each interaction with the algorithm. This alternative can eliminate the need to choose a fixed number for the k -nearest-neighbours.

Note that Eq. (14) for estimating the y -values is based exclusively on the Euclidean distances within an $\mathbf{X}$ space, which can be calculated even if the number of components is greater than the number of objects already analysed.

$$\hat{y}_j = \frac{\sum_{ind \in N_{k(k, i_{sel})}} \frac{y_{ind}}{\|x_{ind} - x_j\|}}{\sum_{ind \in N_{k(k, i_{sel})}} \frac{1}{\|x_{ind} - x_j\|}}, \quad j \in i_{sel}^- \quad (14)$$

Algorithm 1 Active Searching algorithm using kNNR

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1: procedure kNNR( $n_0, n_{max}, i_{sel}, i_{sel}^-, \mathbf{D}, \mathbf{X}, y, freq$ )
2:    $n \leftarrow n_0$   $\triangleright n_0$  denotes the indices selected by the KS algorithm
3:   repeat  $\triangleright$  the block keeps iterating until it reaches the
       maximum number of objects
4:      $k \leftarrow n$   $\triangleright k$ -nearest neighbour
5:     for  $j = 1, \dots, N$  do  $\triangleright j$  is the  $j$ -th  $i_{sel}^-$ 
6:        $\hat{y}_j = \frac{\sum_{ind \in N_{k(k, i_{sel})}} \frac{y_{ind}}{\|x_{ind} - x_j\|}}{\sum_{ind \in N_{k(k, i_{sel})}} \frac{1}{\|x_{ind} - x_j\|}}, ind_j \in i_{sel}^-$   $\triangleright$  estimates the
       property of interest
7:     end for
8:     if ( $59.5 \leq freq_i \leq 60$ ) then  $\triangleright$  bluetests if frequency is within
       operational limits
9:        $ind_{n+1} \leftarrow \arg \max_{i \in \{1, 2, \dots, N\}} f(i)$ 
10:    end if
11:     $i_{sel} \leftarrow i_{sel} \cup ind_{n+1}$ 
12:     $i_{sel}^- \leftarrow i_{sel}^- \setminus ind_{n+1}$   $\triangleright$  update the index sets
13:     $n \leftarrow n + 1$ 
14:  until  $n \leq n_{max}$ 
15:  return  $i_{sel}$   $\triangleright$  returns the selected indices
16: end procedure

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At line 6, the algorithm estimates the values y that correspond to the total loss (the property of interest). At line 8, a test is performed to verify whether the frequency meets the established criteria. This condition ensures that all clusters operate within the allowed frequency range of 59.5 Hz to 60 Hz.

This property is advantageous over multiple linear regression methods, which usually require variable selection techniques when the number of calibration objects is small. Besides, there is no need to choose a related parameter for complex models, such as the number of

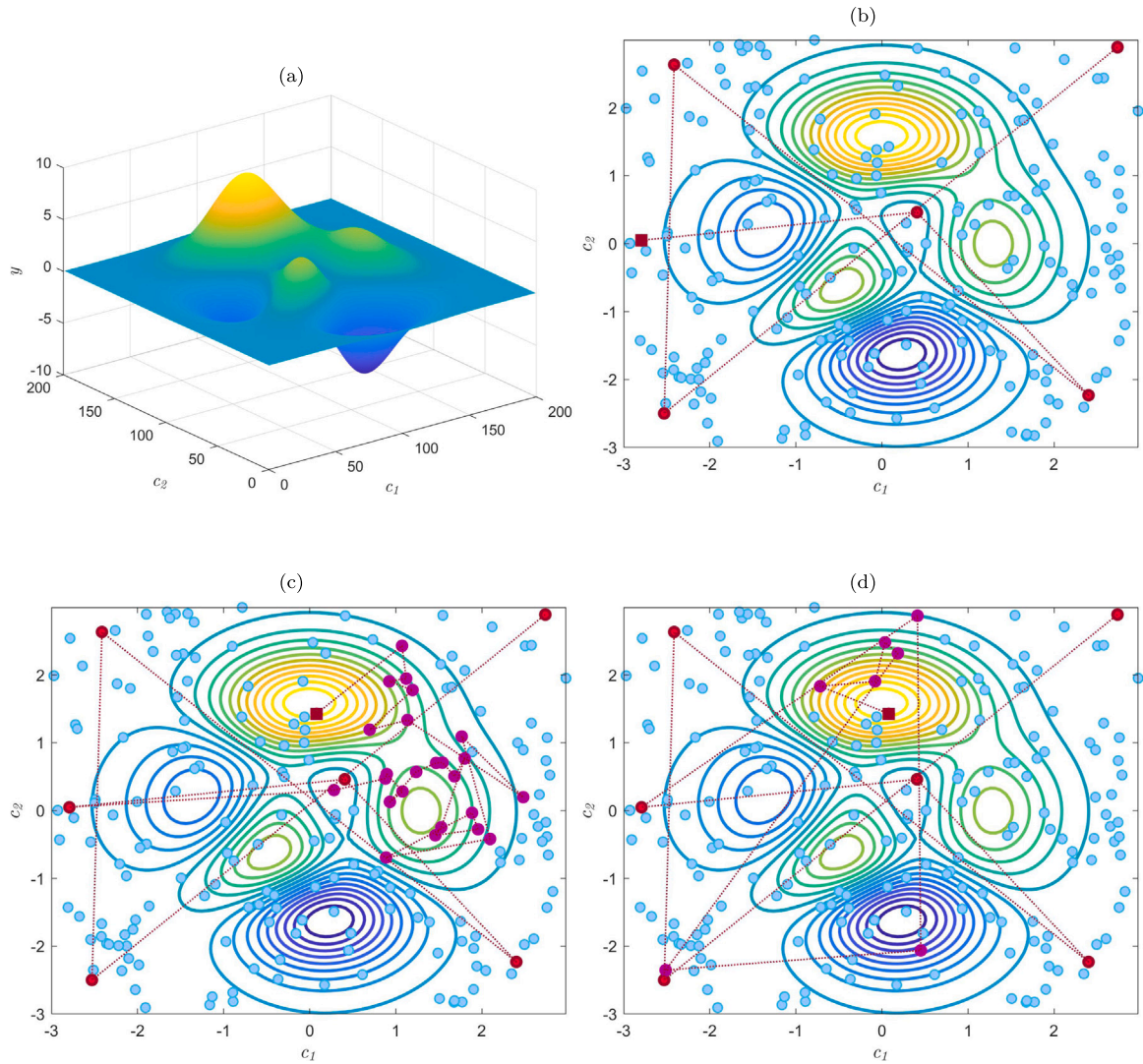


Fig. 2. Results of the active search method in the simulated example: (a) equation graph (b) 6 initial objects selected by the KS algorithm; (c) optimum point after selecting 31 objects using kNNR; (d) optimum point found in the 14th object evaluated using GPR.

Partial Least Squares (PLS) modelling factors. This technique reduces predictors to a smaller set of uncorrelated components and performs least squares regression on these components, instead of on the original data.

3.3.3. Gaussian process regression

GPR can be seen as a Bayesian nonparametric approach to regression, where the function from the Gaussian processes takes values in a function space [11]. As in the previous case, it is assumed that the initial objects were selected using the KS algorithm and that a total of $n \leq n_{max}$ objects are selected at the end of the active search.

As the active search aims to improve the accuracy, remaining sensitive to the optimization costs. Thus, it was necessary to define a criterion by calculating the confidence intervals for the estimated values $\hat{y}$. After the training, the prediction of the $\hat{y}$ -values for the unselected objects is made, and the index of the object with the highest value of $\hat{y}$ is selected. Next, the confidence interval is tested. If the algorithm does not find a confidence interval value greater than the y -value already selected, the algorithm is terminated. For this study, the 95% confidence interval was considered

For the GPR model, we adopted the squared exponential kernel with Automatic Relevance Determination (ARD), as implemented in MATLAB® (“ardsquaredexponential”). This kernel automatically tunes

the length-scale hyperparameters for each input dimension, enhancing model flexibility and precision in high-dimensional spaces. Hyperparameters were optimized using the log marginal likelihood maximization method available in the MATLAB regression learner. Additionally, to reduce computational burden during iterative retraining, we implemented a stopping criterion based on the confidence interval of predictions: if the upper bound of the confidence interval for a candidate configuration was lower than the best observed value, the search was terminated early.

3.3.4. Simulated example

The following simulated example (Fig. 2) illustrates the effectiveness of the proposed search method. The 200 objects involved in this example are indicated in a contour plot of the function associated with the y -values. This objects are created with a random number generator using the factory Mersenne Twister, with seed 0. This example involves two components (c_1 and c_2) related to a property y according to Eq. (15). This expression generates a surface with local and global maxima, as can be seen in Fig. 2(a).

$$y = 3(1 - c_1)^2 e^{-c_1^2 - (c_1 + 1)^2} - 10\left(\frac{c_1}{5} - c_1^3 - c_2^5\right) e^{-c_1^2 - c_2^2} - \frac{1}{3} e^{-(c_1 + 1)^2 - c_2^2} \quad (15)$$

The n_0 objects selected by KS are plotted with red markers connected by dotted lines, for this example there are 6 initial objects, as shown in Fig. 2(b).

From the n_0 objects selected, the application of the kNNR method led to the determination of the highest value of y , in the selection of the 31th object, as shown in Fig. 2(c).

Fig. 2(d) shows that in the 13th selection the object with the highest value for y is found, using active search with the GPR technique. As can be seen, the maximum value of y is reached quickly by the active search algorithm with kNNR and GPR after evaluating $n = 31$ and $n = 14$ objects respectively.

The initial selection favoured an exploration of x -space, with objects scattered throughout the c_1 - c_2 plane. Note that the algorithm reduced the search space during the active search, narrowing the focus to a smaller region during the exploration phase. In this simulation, the GPR quickly converged to the optimal point of the function.

4. Simulation results and discussions

This paper uses active search to determining a near-optimal grouping for an island microgrid network aimed at maximizing service capacity. The network is divided into six microgrids (Fig. 1) in which active search studies are carried out. Each microgrid corresponds to a standard radial distribution system. Depending on the connections available, the microgrids can be grouped, forming multiple clusters, which, in turn, can be further arranged in cluster groups, as shown in Table 1. In normal operation, the microgrids are connected to the main system. For the simulations carried out, it is assumed that the microgrids are isolated from the main electricity system as a function of a disconnection from the main grid, and disconnection time is known. Also, the maximum power supplied is known, which is considered by the methodology to manage the energy supply during the island mode of operation.

This work presents dispatchable energy sources on Buses 725, 729, 731, 225, 229 and 231. On the other hand, non-dispatchable units have variable energy generation since the energy source is intermittent and dependent on weather conditions. The main examples of non-dispatchable units are wind and solar generation. This work uses only photovoltaic generation as an intermittent energy source at Buses 724, 742, 224, and 242. Notice that Microgrids 3 and 4 only have dispatchable energy sources.

The disconnection time directly affects the management of the available energy and, ultimately, the system's ability to meet the load demand. The proposed algorithm is designed to account for this characteristic. In this context, islands are formed based on all possible combinations of the previously created mini-microgrids. The identification of near-optimal groupings is performed by the active search algorithm.

The study considers the disconnection of the microgrids from the main system, and each microgrid's predicted energy generation is assumed. The base power adopted is 2.5 kVA, so a generation of, for example, 2 p.u.h corresponds to 5 kWh.

4.1. Proposed framework

The proposed framework, as can be seen in Fig. 3, consists of 4 phases: (1) specification of all possible microgrid groupings, (2) definition of the problem components, (3) correlation analysis and (4) application of the active search algorithm to find the near-optimal grouping of microgrids.

Phase 1: In this phase, the six islanded microgrids (Table 1) can be combined into 32 different groupings. In this table, the first column indicates the grouping number, while the subsequent columns correspond to clusters 1 through 6. It should be noted that each grouping consists of one or more clusters (up to six), and each cluster may contain one or more microgrids (up to six). For example, Grouping 1 represents a single cluster containing all six microgrids.

Phase 2: Each grouping described in Table 1 corresponds to an object, which is made of components: the energy generated and the

energy demanded by each cluster. For a better understanding, Table 2 (ID 2) describes an example of object configuration, which relates to Grouping 2.

Phase 3: The correlation of the components was analysed and the components with high correlation are removed. Thus, the components c_2 , c_4 , c_6 , c_8 , c_{10} and c_{12} were removed from the objects, leaving 5 components, c_1 , c_3 , c_5 , c_7 , c_9 and c_{11} which correspond to the energy generated from the clusters 1–6.

Phase 4: Active search is used to find the grouping with the highest energy availability to serve the loads when the system is isolated, i.e., the grouping with the lowest total loss, with all clusters operating within the allowed frequency range ($59.5 \text{ Hz} \leq \text{freq} \leq 60 \text{ Hz}$). In the simulations, calculating the total loss of each grouping requires solving the power flow during the period of disconnection for each cluster within the grouping: the total loss corresponds to the sum of the losses of each cluster.

4.2. Defining the near-optimal microgrid clustering for islanded operation

The initial number of training objects (groupings) was determined using the KS algorithm. A value of $n_0 = 2$ (initial objects) was chosen. The values of the property of interest (y), which correspond to the total loss, are then computed for the two selected objects by solving the power flow for each of the corresponding groupings. It is important to note that power flow calculations are not performed for the remaining objects. Additionally, the frequency associated with each cluster of these groupings is derived from the solution of the power flow problem. These objects are subsequently used to train the regression model (kNNR or GPR).

Next, the adjusted regression model is applied to calculate $\hat{y}$ (estimated value for property of interest) for the 30 remaining objects. The object with the highest $\hat{y}$ is selected, and its corresponding y value is calculated, meaning that the power flow calculation is performed for this object (or grouping). This object is then incorporated into the training set of the regression model.

In the third iteration, the regression model is updated with the three objects (those for which the power flow was calculated in previous iterations) to estimate $\hat{y}$ for the remaining 29 objects. This process is repeated until a stopping criterion is met. The stopping criterion is determined by calculating the confidence intervals for the estimated $\hat{y}$ values. For this study, a 95% confidence interval was considered. If the $\hat{y}$ interval does not exceed the y values of the objects already selected, the object is added to the training set; otherwise, the algorithm terminates. This criterion was met after the ninth iteration, as indicated by the red dashed line, as shown in Figs. 4(a) and 4(b). In this case, the active search method for determining microgrid groupings quickly identified the near-optimal grouping. In contrast, when no search method is used, the near-optimal grouping is only identified in the 22nd iteration.

Fig. 4(a) compares the kNNR active search (kNNR AS) method with the no-search method. The kNNR AS technique (as outlined in Algorithm 1) is then used to estimate the total loss for each microgrid grouping that has not yet been selected. The grouping with the lowest estimated total loss, which also satisfies the defined constraint, is subsequently chosen. In this case, this was identified during the third iteration. The GPR active search (GPR AS) is then used to estimate the total loss for each microgrid grouping that have not yet been selected, similar to kNNR AS. In this case, this microgrid grouping with the lowest total was found during the third interaction. Fig. 4(b) compares the results obtained for active search using the GPR technique and no-search method.

The grouping with the least energy loss was identified by the active search method when evaluating the third object using both the kNNR AS and GPR AS techniques. This grouping corresponds to object 22. In this study, the better solution was reached after only three iterations.

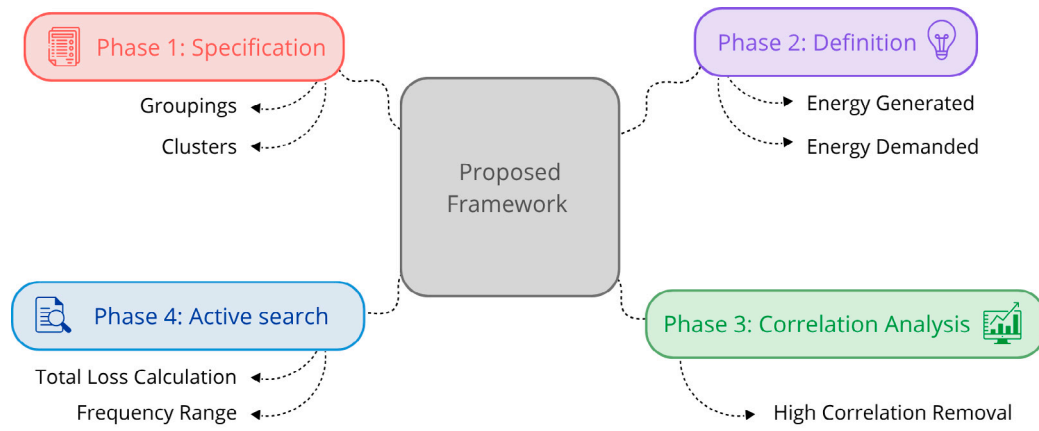


Fig. 3. Proposed framework.

Table 1
Microgrid clusters in the proposed system.

ID	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6
01	M1-M2-M3-M4-M5-M6					
02	M1-M2-M3-M4-M5	M6				
03	M1-M2-M3-M4	M5-M6				
04	M1-M2-M3-M4	M5	M6			
05	M1-M2-M3	M4-M5-M6				
06	M1-M2-M3	M4-M5	M6			
07	M1-M2-M3	M4	M5-M6			
08	M1-M2-M3	M4	M5	M6		
09	M1-M2	M3-M4-M5-M6				
10	M1-M2	M3-M4-M5	M6			
11	M1-M2	M3-M4	M5-M6			
12	M1-M2	M3-M4	M5	M6		
13	M1-M2	M3	M4-M5-M6			
14	M1-M2	M3	M4-M5	M6		
15	M1-M2	M3	M4	M5-M6		
16	M1-M2	M3	M4	M5	M6	
17	M1	M2-M3-M4-M5-M6				
18	M1	M2-M3-M4-M5	M6			
19	M1	M2-M3-M4	M5-M6			
20	M1	M2-M3-M4	M5	M6		
21	M1	M2-M3	M4-M5-M6			
22	M1	M2-M3	M4-M5	M6		
23	M1	M2-M3	M4	M5-M6		
24	M1	M2-M3	M4	M5	M6	
25	M1	M2	M3-M4-M5-M6			
26	M1	M2	M3-M4-M5	M6		
27	M1	M2	M3-M4	M5-M6		
28	M1	M2	M3-M4	M5	M6	
29	M1	M2	M3	M4-M5-M6		
30	M1	M2	M3	M4-M5	M6	
31	M1	M2	M3	M4	M5-M6	
32	M1	M2	M3	M4	M5	M6

Table 2
Object 2 configuration related to Grouping 2.

Cluster 1: M1-M2-M3-M4-M5 — cluster 2: M6 — cluster 3-6: empty
c_1 : energy generated - cluster 1
c_2 : energy demanded - cluster 1
c_3 : energy generated - cluster 2
c_4 : energy demanded - cluster 2
c_5 : 0 (energy generated - cluster 3)
c_6 : 0 (energy demanded - cluster 3)
c_7 : 0 (energy generated - cluster 4)
c_8 : 0 (energy demanded - cluster 4)
c_9 : 0 (energy generated - cluster 5)
c_{10} : 0 (energy demanded - cluster 5)
c_{11} : 0 (energy generated - cluster 6)
c_{12} : 0 (energy demanded - cluster 6)

This approach required just 9.4% of the total power flow calculations that would be needed in a continuous analysis, significantly streamlining the process.

The active search method significantly reduced the number of iterations required to select the near-optimal configuration that meets the load demand within the established frequency, while minimizing total loss. It is important to note that the value of the property of interest, in the case of microgrids, was achieved after a limited number of iterations, thereby reducing the number of control system operations needed to execute the power flow calculations.

The selected near-optimal configuration (Grouping 22) explicitly satisfied all specified operational constraints (frequency stability and voltage limits) verified through detailed power flow simulations. Thus, this configuration not only minimizes total active power losses but also ensures compliance with critical operational requirements. In this configuration, Microgrids 1 and 6 are isolated, while Microgrids 2–3 and 4–5 are grouped together.

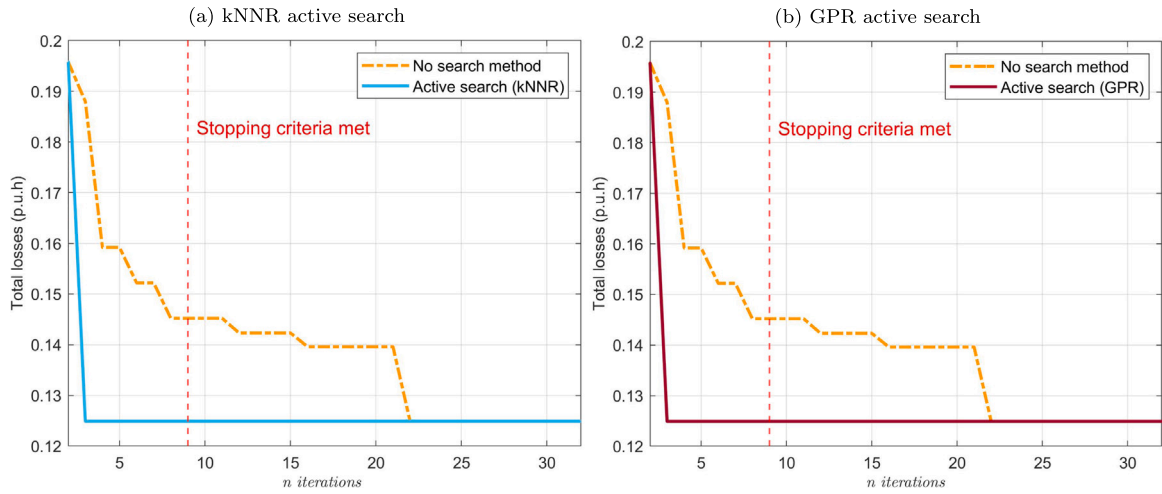


Fig. 4. Result of the active search method for choosing the near-optimal microgrid.

While the initialization of kNNR and GPR is essentially similar, the main differences and advantages lie in the model training. The basic principle and process of kNNR are not considered complex, as they rely on a distance-based criterion to estimate the value of y . On the other hand, GPR incorporates a Bayesian model during the training process, providing a probabilistic framework for making predictions and quantifying uncertainty.

In the kNNR technique, the parameter k used to estimate the property y is a design variable that must be selected by the researcher. Good results were also obtained using a simpler version of this technique, in which all the selected objects are used to estimate y . In contrast, GPR is highly dependent on the hyperparameters of its kernel function, which vary according to the nature of the data, making the algorithm more adaptable than kNNR.

The time to find the best scenario using kNNR AS was 0.106 ms and using GPR AS was 1.643 ms, within a MATLAB® programming environment. The tests were carried out on a computer with an Intel® Core i7-6700, Central Processing Unit (CPU) 3.41 GHz, with 4 GB of Random Access Memory (RAM) installed, using the Windows® 10 Education operating system.

Metaheuristic optimization techniques, such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) are widely used for constrained optimization problems, including microgrid planning and reconfiguration. However, they typically require a large number of objective function evaluations to converge. When applied to scenarios that involve repeated power flow simulations, these techniques can become computationally prohibitive, especially during planning stages or in real-time applications. In contrast, the basic principle and process of kNNR AS are not considered complex, as they rely on a distance-based criterion to estimate the value of y . The number k used to estimate the property y is a design parameter that must be chosen by the researcher.

The GPR incorporates a Bayesian model during the training process, providing a probabilistic framework for making predictions and quantifying uncertainty. GPR is highly dependent on the hyperparameters of its kernel function, which vary according to the nature of the data, making the algorithm more adaptable than kNNR.

Although the experimental validation in this study focuses on comparisons with the KS selection method, it is important to contextualize the proposed methodology in relation to other clustering and optimization techniques commonly found in the literature. To this end, Table 3 presents a comparative summary of different active learning methods, including PSO and GA. The comparison encompasses aspects such as model complexity, parameter tuning, interpretability, adaptability, average number of iterations, and source of information. The data

for PSO and GA are drawn from published studies addressing similar microgrid optimization scenarios.

The GA demonstrates strong performance in addressing high-dimensional and nonlinear optimization problems, due to its population-based evolutionary structure, which promotes a broad search space exploration and reduces the likelihood of being trapped in local optima. This makes GA particularly effective for the optimization of hybrid microgrid systems that integrate multiple distributed energy resources and require simultaneous consideration of environmental and economic objectives [21,22]. In contrast, PSO operates through relatively simple update mechanisms involving particle velocity and position, which results in a lower computational burden compared to GA [23]. However, advanced PSO variants, particularly those integrating adaptive inertia weights or constriction factors, have exhibited superior ability to balance exploration and exploitation. Consequently, these approaches attain performance levels comparable to GA in solving complex optimization problems, all while preserving higher convergence efficiency [24].

Therefore, although this study does not offer an exhaustive numerical comparison with all existing methods, the proposed approach demonstrates strong advantages in terms of computational efficiency, robustness, and practical applicability in constrained power system optimization problems.

In this paper, we systematically explain for the first time how active learning can be applied to determining near-optimal groupings of islanded microgrids. The results of these studies demonstrate that the active search method holds significant potential for addressing this problem.

5. Conclusion

This work developed a structured approach to determine a near-optimal grouping for an islanded microgrid network, aiming to maximize service capacity through an active-learning paradigm and minimize the number of simulations.

Primary control was employed based on the droop method, and an islanding technique has been proposed to optimize the system configuration so that the load supply is maximized while preserving the operating constraints. The methodology proposed may provide dynamic islanding formation and manage the energy available as a function of the disconnection time.

This article proposes an innovative approach and presents the following key findings. First, the study successfully applies microgrid clustering to enhance service capacity during disconnection from the main electricity grid. Second, the active-search method significantly reduces the number of iterations required to determine the near-optimal

Table 3

Comparison of the proposed active search approach with typical metaheuristic techniques for microgrid optimization.

Characteristic	kNNR AS	GPR AS	PSO	GA
Model complexity	Low complexity.	High complexity.	Moderate	Moderate
Parameter tuning	Requires tuning the number of neighbours.	Requires selection of the kernel and optimization of hyperparameters.	Requires multiple parameters (inertia weight, cognitive, social) and inertial adaptive factors can be automatically adjusted	Requires tuning multiple operators: population size (100), crossover and mutation rates
Interpretability	Easy to interpret and understand.	More difficult to interpret due to the probabilistic nature.	Easy to interpret and understand.	Easy to interpret and understand.
Adaptability	Limited adaptability to complex relationships.	High adaptability, can model complex, nonlinear relationships.	Adapts well to multi-objective problems	High adaptability
Iteration	~3	~3	~200	~200
Source	This work	This work	[24]	[21,23,24]

grouping of microgrids, demonstrating its efficiency. Finally, the proposed framework offers a systematic approach to optimizing energy distribution while maintaining operational constraints.

The active search yielded promising results in selecting the near-optimal configuration to meet load demand requirements within the specified frequency. Two regression techniques were investigated: KNNR and GPR. The latter is more promising due to its ability to learn both the mean and covariance functions from the training data, enabling it to make predictions for new data points and represent uncertainty distributions through Gaussian processes.

5.1. Limitations of the current work and suggestions for future investigation

While the proposed method efficiently identifies high-performance, practically viable solutions, it does not guarantee global optimality due to its reliance on predictive surrogate modelling. Thus, the results are appropriately categorized as near-optimal, well-suited for practical microgrid operational planning where computational efficiency and quick response are priorities. Despite the method's demonstrating high efficiency and effectiveness in identifying near-optimal microgrid groupings under islanded conditions, it is important to acknowledge some limitations:

- No optimized hyperparameters of the Gaussian process regression: The difficulty in configuring these hyperparameters stems from the wide range of possibilities they offer. For future work, it is recommended to explore methods to optimize these parameters.
- No real-time reconfiguration: The method is designed for pre-planning, not real-time operation. Integrating it with real-time control systems would require additional layers of decision-making and validation.
- No explicit failure scenarios: The method assumes full availability of the microgrid infrastructure. Modelling grid component failures (e.g., line faults, inverter shutdowns) would enhance the robustness of the optimization strategy.

CRedit authorship contribution statement

Cláudia Eliane da Matta: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Eliane Valença do Nascimento De Lorenci:** Writing – review & editing, Validation, Resources, Methodology, Investigation, Conceptualization. **João Alves da Silva Neto:** Writing – review & editing, Validation, Software, Resources. **Antonio Carlos Zambroni de Souza:** Writing – review & editing, Supervision. **Pedro Paulo Balestrassi:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] N.F.P. Dinata, M.A.M. Ramli, M.I. Jambak, M.A.B. Sidik, M.M. Alqahtani, Designing an optimal microgrid control system using deep reinforcement learning: A systematic review, *Eng. Sci. Technol. an Int. J.* 51 (2024) 101651, <http://dx.doi.org/10.1016/j.jestch.2024.101651>.
- [2] F. Calero, C.A. Cañizares, K. Bhattacharya, C. Anierobi, I. Calero, M.F.Z. de Souza, M. Farrokhhabadi, N.S. Guzman, W. Mendieta, D. Peralta, B.V. Solanki, N. Padmanabhan, W. Violante, A review of modeling and applications of energy storage systems in power grids, *Proc. IEEE* 111 (7) (2023) 806–831, <http://dx.doi.org/10.1109/JPROC.2022.3158607>.
- [3] R.F. Alonso, A.C. Zambroni de Souza, Resilience of active networks with optimal mobile energy storage systems management, *Electr. Power Syst. Res.* 217 (2023) 109152, <http://dx.doi.org/10.1016/j.epsr.2023.109152>.
- [4] Y. Ding, J.D. Martín-Guerrero, M. Sanz, R. Magdalena-Benedicto, X. Chen, E. Solano, Retrieving quantum information with active learning, *Phys. Rev. Lett.* 124 (2020) 140504, <http://dx.doi.org/10.1103/PhysRevLett.124.140504>.
- [5] R. Gelbhart, R. El-Yaniv, The relationship between agnostic selective classification, active learning and the disagreement coefficient, *J. Mach. Learn. Res.* 20 (1) (2019) 1136–1173, doi:<https://dl.acm.org/doi/10.5555/3322706.3361974>.
- [6] A. Krishnamurthy, A. Agarwal, T.-K. Huang, H. Daumé, J. Langford, Active learning for cost-sensitive classification, *J. Mach. Learn. Res.* 20 (1) (2019) 2334–2383, URL <http://jmlr.org/papers/v20/17-681.html>.
- [7] A. Ukil, J. Bernasconi, Neural network-based active learning in multivariate calibration, *IEEE Trans. Syst. Man, Cybern. Part C (Appl. Rev.)* 42 (6) (2012) 1763–1771, <http://dx.doi.org/10.1109/TSMCC.2012.2220963>.
- [8] T. Cover, P. Hart, Nearest neighbor pattern classification, *IEEE Trans. Inform. Theory* 13 (1) (1967) 21–27, <http://dx.doi.org/10.1109/TIT.1967.1053964>.
- [9] C.E. Rasmussen, H. Nickisch, Gaussian processes for machine learning (GPML) toolbox, *J. Mach. Learn. Res.* 11 (2010) 3011–3015, doi:<https://dl.acm.org/doi/10.5555/1756006.1953029>.
- [10] A.C. de Souza, M. Santos, M. Castilla, J. Miret, L. de Vicuña, D. Marujo, Voltage security in AC microgrids: a power flow-based approach considering droop-controlled inverters, *IET Renew. Power Gener.* 9 (8) (2015) 954–960, <http://dx.doi.org/10.1049/iet-rpg.2014.0406>.
- [11] C. Ballabio, E. Lugato, O. Fernandez-Ugalde, A. Orgiazzi, A. Jones, P. Borrelli, L. Montanarella, P. Panagos, Mapping LUCAS topsoil chemical properties at European scale using Gaussian process regression, *Geoderma* 355 (2019) <http://dx.doi.org/10.1016/j.geoderma.2019.113912>.
- [12] F. Mumtaz, M.H. Syed, M.A. Hosani, H.H. Zeineldin, A novel approach to solve power flow for islanded microgrids using modified Newton-raphson with droop control of DG, *IEEE Trans. Sustain. Energy* 7 (2) (2016) 493–503, <http://dx.doi.org/10.1109/TSTE.2015.2502482>.

- [13] P.J. Lagace, M.H. Vuong, I. Kamwa, Improving power flow convergence by Newton Raphson with a Levenberg–Marquardt method, in: 2008 IEEE Power and Energy Society General Meeting - Conversion and Delivery of Electrical Energy in the 21st Century, IEEE, 2008, pp. 1–6, <http://dx.doi.org/10.1109/PES.2008.4596138>.
- [14] B. de Nadai Nascimento, A.C. de Souza, J.G. de Carvalho Costa, M. Castilla, Load shedding scheme with under-frequency and undervoltage corrective actions to supply high priority loads in islanded microgrids, IET Renew. Power Gener. 13 (11) (2019) 1981–1989, <http://dx.doi.org/10.1049/iet-rpg.2018.6229>.
- [15] IEEE standard for interconnection and interoperability of distributed energy resources with associated electric power systems interfaces, IEEE Std 1547-2018 (Revis. IEEE Std 1547-2003) (2018) 1–138, <http://dx.doi.org/10.1109/IEEESTD.2018.8332112>.
- [16] C.E. Matta, H.M. Paiva, R.K.H. Galvao, M.C.U. Araujo, S.F.C. Soares, K.C. Weber, L.A. Pinto, An active search method for finding objects with near-optimal property values within a given set, J. Braz. Chem. Soc. 27 (7) (2016) 1177–1187, <http://dx.doi.org/10.5935/0103-5053.20160014>.
- [17] G. dal Bianco, M.A. Goncalves, D. Duarte, BLOSS: Effective meta-blocking with almost no effort, Inf. Syst. 75 (2018) 75–89, <http://dx.doi.org/10.1016/j.is.2018.02.005>.
- [18] R.W. Kennard, L.A. Stone, Computer aided design of experiments, Technometrics 11 (1) (1969) 137–148, <http://dx.doi.org/10.2307/1266770>.
- [19] M. Durbin, M.A. Wonders, M. Alaska, A.T. Lintereur, K-nearest neighbors regression for the discrimination of gamma rays and neutrons in organic scintillators, Nucl. Instrum. Methods Phys. Res. Sect. A: Accel. Spectrom. Detect. Assoc. Equip. 987 (2021) 164826, <http://dx.doi.org/10.1016/j.nima.2020.164826>.
- [20] D. Wang, X. Tan, Bayesian neighborhood component analysis, IEEE Trans. Neural Netw. Learn. Syst. 29 (7) (2018) 3140–3151, <http://dx.doi.org/10.1109/TNNLS.2017.2712823>.
- [21] Y. Zhang, et al., Sizing of hybrid PV/battery/wind/diesel microgrid system using an improved decomposition multi-objective evolutionary algorithm considering uncertainties and battery degradation, Sustainability 15 (14) (2021) 11073, <http://dx.doi.org/10.3390/su151411073>.
- [22] X. Zhu, G. Ruan, H. Geng, H. Liu, M. Bai, C. Peng, Multi-objective sizing optimization method of microgrid considering cost and carbon emissions, IEEE Trans. Ind. Appl. 60 (4) (2024) 5565–5576, <http://dx.doi.org/10.1109/TIA.2024.3395570>.
- [23] Z. Guan, H. Wang, Z. Li, X. Luo, X. Yang, J. Fang, Q. Zhao, Multi-objective optimal scheduling of microgrids based on improved particle swarm algorithm, Energies 17 (7) (2024) <http://dx.doi.org/10.3390/en17071760>, URL <https://www.mdpi.com/1996-1073/17/7/1760>.
- [24] K. Ghanbari, A. Maleki, D.R. Ochbelagh, Investigating the effect of various types of components in optimal designing of a solar/wind/storage hybrid system, J. Energy Storage 110 (2025) <http://dx.doi.org/10.1016/j.est.2024.115273>.